

BRIGHTLEARN DATA ANALYTICS - RESEARCH ASSIGNMENT - 1

Section A: Data Fundamentals

no. 1 - Database -

it is a collection of information, facts which can be used to analyse, for instance for a business keeping data of day to day operations, sales, client base, enables business owner to determine whether business is growing or not. It helps business owners to know which group age to target for which product and how to keep enticing them to keep buying.

no. 2. Datawarehouse -

It is a big storage which stores complex data collected from regular database.

Difference between - Datawarehouse and Regular database

a) Datawarehouse

- Collects and stores complex data into categorised divisions
- It summarises data which will later enable data analyst, engineer and data scientist to be able to do their work.
- ~~BI is easy~~: Enables BI (business Intelligence) for an organisation or an enterprise to make informed decisions. ~~like~~

b) Regular database -

- Records / or keeps record of daily operations data uncategorised.
- It updates daily / frequently as transactions and operation take place.
- ~~On the other~~ whereas datawarehouse updates on a set scheduled time frame only.

Why a company would use data warehouse

- It makes it easier for a business to analyse complex data throughout all organisation departments, and be able to perform BI.

NO. 3 Data Lake

It is a central data storage that stores both structured and ~~as~~ unstructured data. It enables organisation or an enterprise to scale data of any size at a faster pace. It saves organisation time from going through data defining structures, schema and transformations.

Data Lake vs Data Warehouse

ASPECT	Data Lake	Data Warehouse
	Store everything, figure it out later	Store clean stuff, ready reports
Data type	Raw: CSV, JSON, images, video, logs, audio. Any format.	Structured: tables + columns only. Clean
Schema	Schema-on-read. Dump data now, decide structure when you query.	Schema-on-write. Plan + structure it before loading
Query Speed	Slow BI. fast for big ML jobs.	Very fast for dashboards "Sales last month"

No. 4 Data Mart

Data mart is a curated data base which provides a simplified, Subject-oriented view of data. It makes it easier for ~~user~~ user to access information they need promptly and carry out analysis ~~at~~ without long delay by navigating the entire data warehouse.

a) How it (Data mart) relate to data warehouse

- It is built from an existing data warehouse.

b) Who typically uses data mart

- Sales departments - to compile quarterly sales reports
- Marketing - to analyse consumer behaviour and pin point which age group is carrying majority of the purchasing power of the organisation.

No. 5 Difference between;

Structured	Semistructured	Unstructured data
- Not flexible	- Zipped files	- Flexible
- Easy access	- Emails	- Fast data collection
- Easy integration	- JSON	- Easy scalability
- Secure data management	- Integration of data	- Hard to analyse
found in data warehouse		
- Names, credit cards, ID, phone numbers, date etc.		- Lack of specialised tools found in data lakes, Docs, video, media files, surveys, webpages

Structured data

- it is stored in a table with rows and columns
- Has consistent format

Semistructured data

- Info that has organisational properties
- It sits between structured and unstructured data

Unstructured data

- it does not have consistent format.

No. 6 Data modeling

it is the process of creating a structured and visual representation of data, which ~~enables~~ helps translate business requirements into organised data structure, which supports accurate analysis and effective ~~des~~ decision-making.

Importance of data modeling when designing Systems

- It provides structured clear framework
- Enables better decision making
- Helps in visualising complex data structures.
- Offers clear road map.
- It acts as a universal language for easier communication between business and technical stakeholders.
- Ensures ~~and~~ consistency in documentation and System designs across organisations.

Two types of data modeling.

1) Physical data modeling - specifies how data is sorted in a database including storage mechanisms, tables structures and indexes.

2) Conceptual Data - provides high level overview on business requirements with technical details.

no. 7 Mining Data

It is a process used to analyse sets of data to uncover or identify useful patterns, trends etc. Uses techniques like algorithms, statistics, machine learning to identify concealed relations in data.

It used in business to factor below

- Purchasing power.
- Identify which age group to target on certain products based on their social media likes. (or search history).
- In healthcare, it enables the doctors to identify which diseases ~~on high~~ high rise ed certain season of the year, which then helps pharmacies know which medication should not run out of stock during that season.

no. 8 Data Management

It is collecting, organising, storing and securing productivity data to be used efficiently to support decision making productivity, compliance and innovation. It is designed to handle the entire life cycle of data, and ~~interreg~~ interpretes process, tools and policies to manage both structured and unstructured data. Supports analytics, reporting etc.

No. 8 Data Management

- it is storing (collecting), organising, processing and securing data to be later used efficiently to support decision making, productivity, compliance and innovation.
- It is designed, to handle the entire life cycle of data, and integrates; processes tools and policies to manage both structured and unstructured data. Supports analytics, reporting etc.

■ MySQL - it is a popular, scalable, user-friendly, open source and free relational database management language (SQL) to store and manipulate data.

■ Provides Security - offers encryption, authentication and compliance features for ~~indus~~ industries like healthcare and financial industries.

■ E-commerce - manages product catalogs, orders and transactions.

■ High performance - optimized for quick data retrieval and handling large datasets

■ Flexibility - supports both SQL and no SQL (via JSON) in the database.

ORACLE - (developed in 1977 by Larry & Bob) it is a relational database management system developed by Oracle Corporation. It is an advanced RDBMS built to manage large volumes of structured data efficiently.

Oracle allows:-

- It allows seamless integration of data or relational and document models.
- It has built-in AI and machine learning features like AI VECTOR SEARCH, enabling advanced analytics and contextual insights.
- It provides high availability through features like ORACLE APPLICATION CLUSTERS (RAC) and data replication across regions. Security is enhanced with tools like SQL FIREWALL to prevent SQL injections attacks and encryption for sensitive information.